

Efficient Dependency Resolution in IFC-aware Decentralized Programming

Steffan Christ Sølvesten and Aslan Askarov

PriSC 2026



Module systems are difficult!

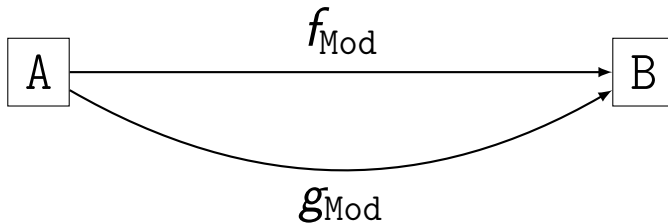
A

B

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How can one add a module cache?

How can one add a *dependency* ~~module~~ cache?

Leaky Cache² (Confidentiality)

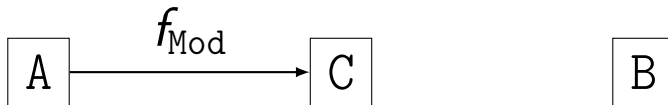
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C

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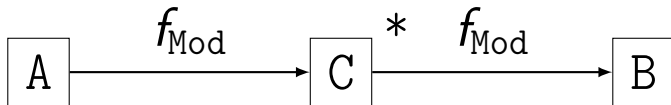
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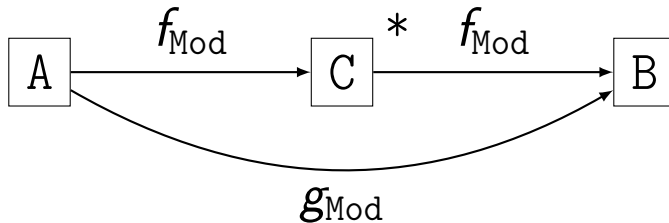
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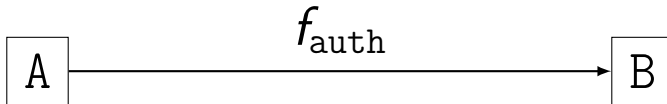
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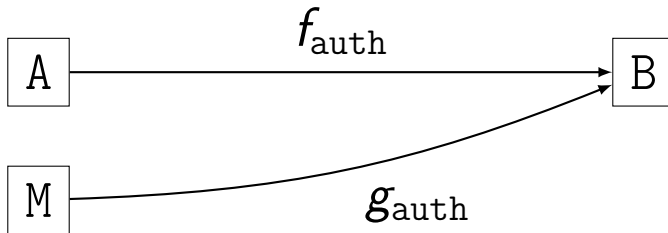
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Replay Dependency³ (Integrity)



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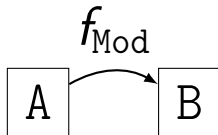
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Chain of Death⁴ (Availability)



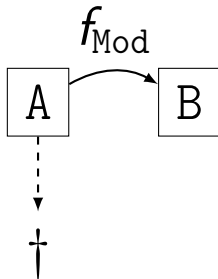
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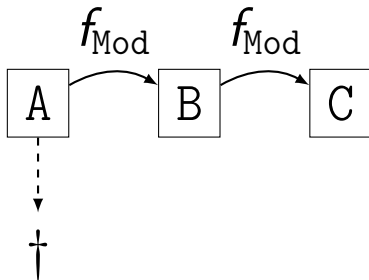
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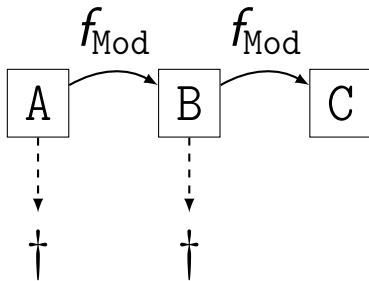
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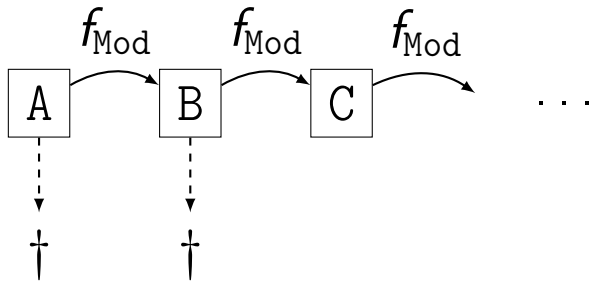
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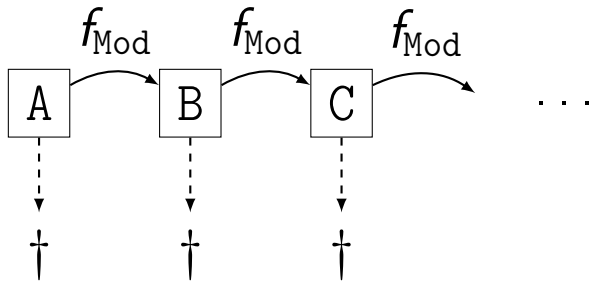
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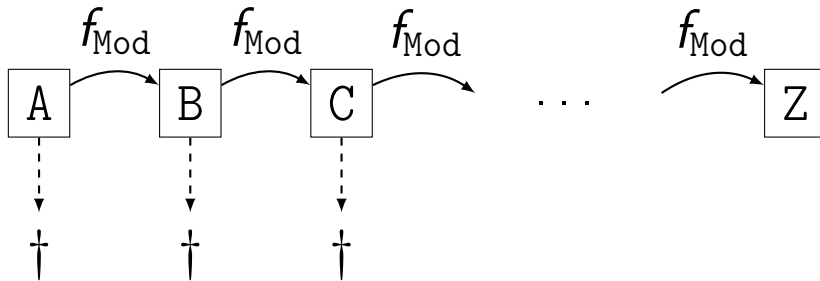
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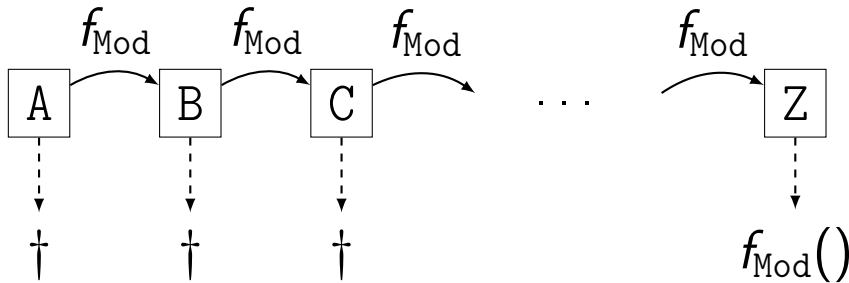
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How can one add an IFC-aware cache?

Troupe Programming Language

```
(* Alice.trp *)
```

```
1 let val msg = "Hello, Bob!"
2     fun f () = print msg
3 in send (@bob, f)
4 end
```

```
(* Bob.trp *)
```

```
5 let fun loop () =
6     receive [hn f => spawn f];
7     loop ()
8 in loop ()
9 end
```

■ Decentralized Programming

For creation of open self-organizing distributed systems.

■ Information Flow Control

Ensures via a monitor that secret information is not by accident shared with nor affected by unwanted parties.

■ Dynamically Typed

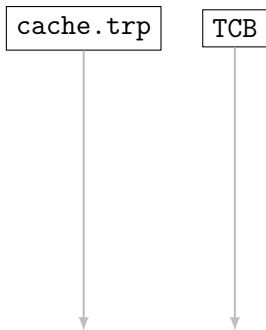
Permits the dynamic behaviour needed by a truly open and updateable system.

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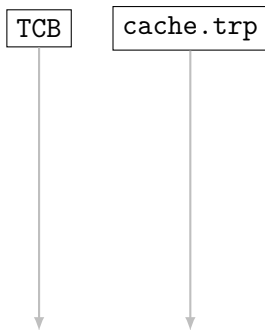
How can one add an IFC-aware cache?
(with the minimal extension to the TCB...)

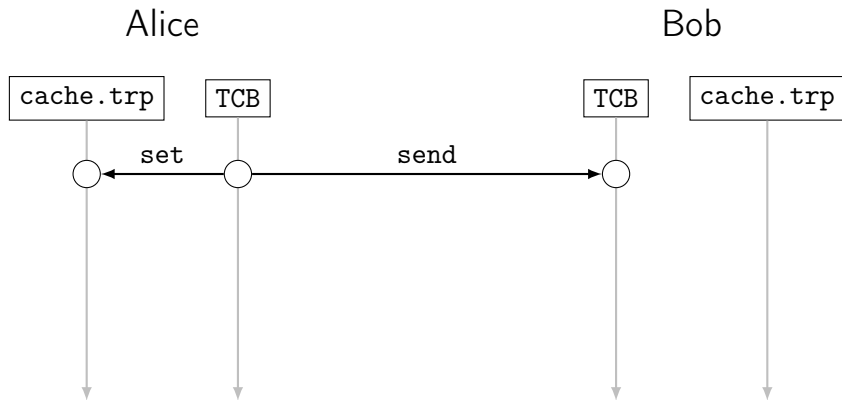
How can one add an IFC-aware cache?
(by using Troupe itself...?)

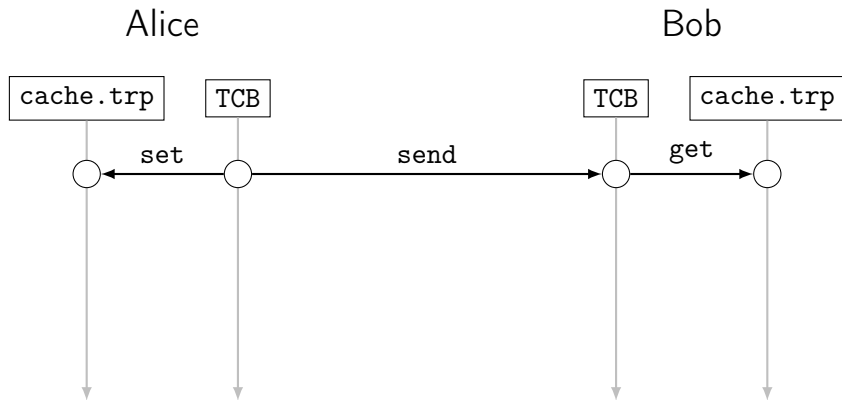
Alice

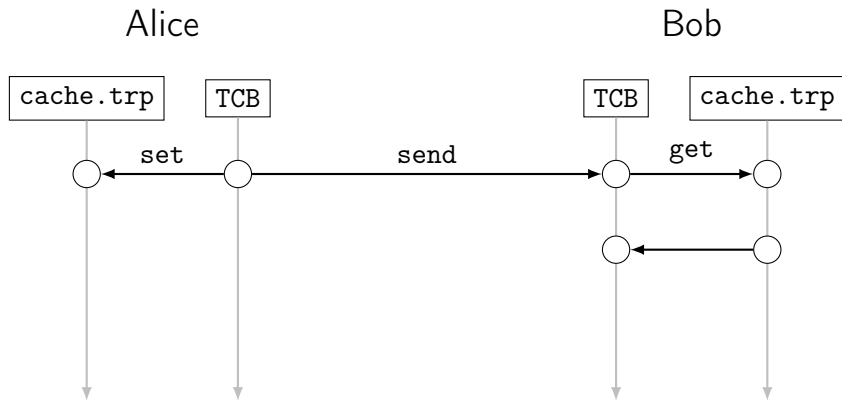


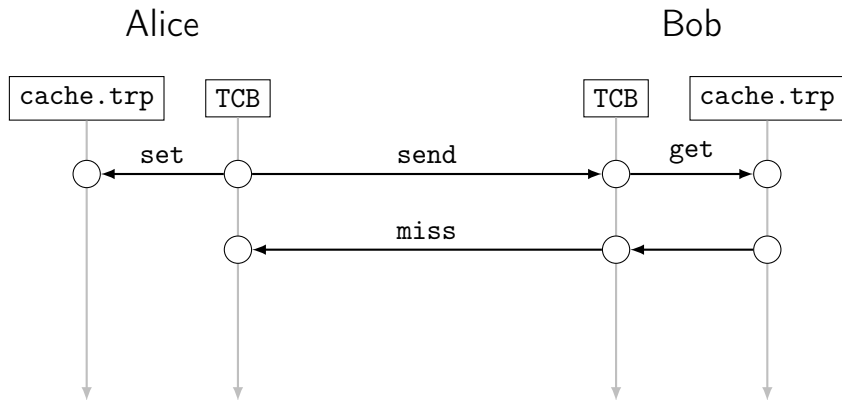
Bob

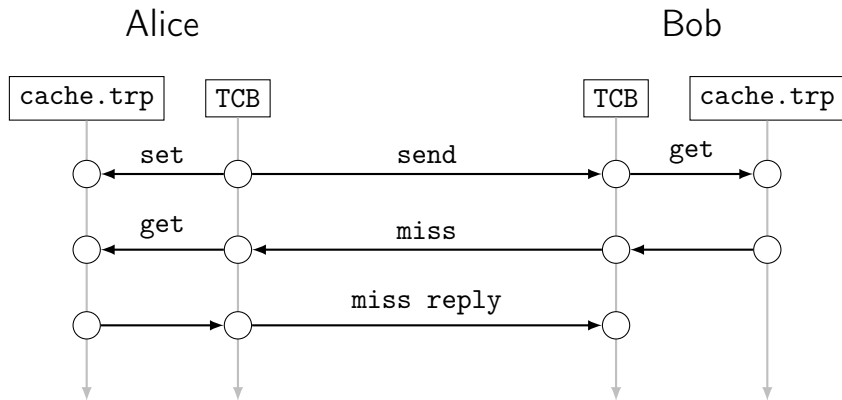


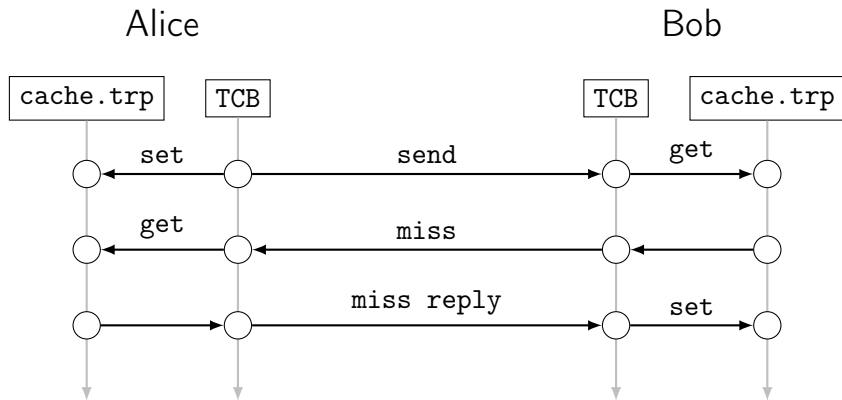












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```
(* cache.trp : line 48 *)
```

```
HashMap.insert m auth k v
```

```
(* cache.trp : line 57 *)
```

```
HashMap.findAll m auth k
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(** API Setter function *)  
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- Use `sandbox builtin` to guard against timing attacks. (14 LOC)

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“Specification”:

An IFC-aware (Hash)Map is a single data structure with public and secret key-value pairs which provides a IFC-aware projections of its insert and remove⁵ history:

$$i(x \mapsto 1)@ \{\}, \quad i(x \mapsto 2)@ \{A\}, \quad i(y \mapsto 3)@ \{B\}, \quad r(x)@ \{B\}, \quad i(y \mapsto 4)@ \{\}, \quad \dots$$

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Changes to Runtime (TCB)

- **Peer-to-peer layer:**

Extend with `miss` and `miss reply` messages which are always sent by value.

- **Serialization:**

Dependency resolution requires introspection of function closures.

- During `serialize`, identify (sub)values to be referenced by hash.
- To `deserialize`, (1) extract references (2) `deserialize` given (all resolved) dependencies.

- **Hash:**

Hashing Troupe values requires introspection of function closures and source.

- **Scheduler:**

Extend to run Troupe code from the peer-to-peer layer.

How can one add an IFC-aware cache?

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By bootstrapping the IFC-aware runtime!

Steffan Christ Sølvsten

✉ soelvsten@cs.au.dk

🌐 ssoelvsten.github.io

Troupe

🔗 github.com/TroupeLang/troupe



Troupe's Threat Model

Uses several security mechanisms, inspired by Jif, Fabric, and FLAM.

- Progress-sensitive Dynamic Information-flow Control
- Sandboxing
- Capabilities

Few assumptions on adversaries! But, we do ignore:

- Network Traffic Analysis. Can be avoided with via recent methods.
(Blaabjerg and Askarov '21 & '23, Nelson, Pagnin, and Askarov '24)
- Timing-channels. Can be avoided via sandboxing and predictive mitigation.
(Zhang, Askarov, Myers '10 & '12)

Additions in Troupe: caching layer (timing-attack mitigation)

Troupe provides the `sandbox(f, t)` operation to run *f* for *t* ms and either return the result, an error, or a continuation. In practice, *t* = 1 ms seems to suffice.

```
(* cache.trp : line 48 *)  
val (m', t') =  
    sandbox' (fn () => HashMap.insert m auth k v) t m  
  
(* cache.trp : line 57 *)  
val (v, t') =  
    sandbox' (fn () => HashMap.findAll m auth k) t []
```

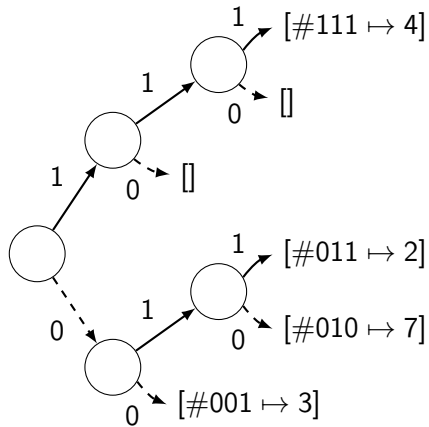

Design of the HashMap

Current prototype uses a Hash Array
Mapped Trie (HAMT) with 200 LOC.

Only 3 declassifications per insert,
remove, find (-24/+55 LOC):

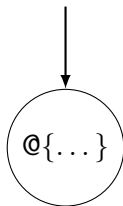
- hash key and hash bucket to not raise pc and taint the entire trie.
- The bucket list (but not its content).

But, this fully declassifies its size!



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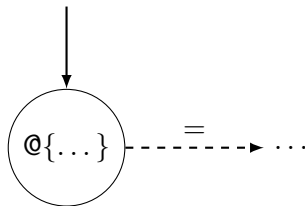
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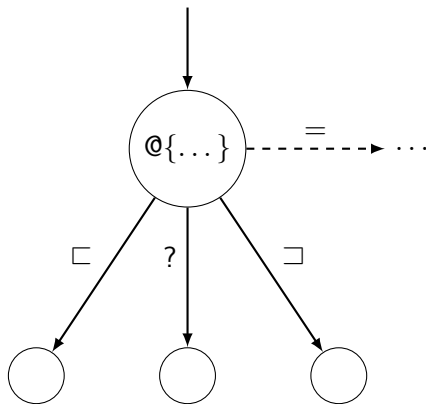
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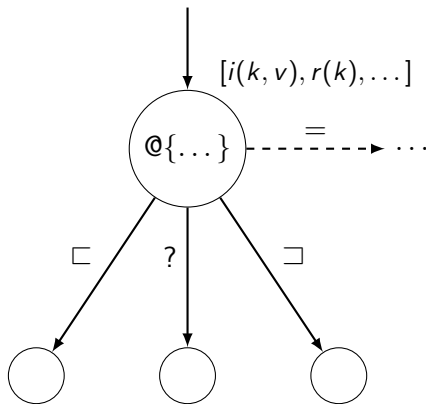
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- Use buffers to accumulate operations for “inaccessible” sub structures. This keeps `insert` and `remove` efficient.



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- Guard taint of secret sub structures with (first-class) IFC labels.
- Use the partial ordering on information flow ($\sqsubseteq$) to compare levels.
- Use buffers to accumulate operations for “inaccessible” sub structures. This keeps `insert` and `remove` efficient.
- Use time-stamps to identify latest key-value from sub structures.

But, does this make `find` too slow?

